

SECTION A (12 marks)

1. Explain the following concepts in Python with short examples
 - a) Polymorphism
 - b) Inheritance
 - c) Encapsulation (6 marks)
2. Write the basic syntax for handling a file in read mode such that the content of the file, “example.txt”, is displayed either entirely or line by line. (2 marks)
3. Give two advantages of logging in programming. (2 marks)
4. Differentiate between sets and lists. (2 marks)

SECTION B (8 marks)

Using your preferred Python IDE, solve the problem below, save the program as a .py file and share your solution via email to the provided address. Comment your name and matricule number in the code, for identification.

Write a Python program that:

1. Defines a **BankAccount** class with:
 - Private Attribute: **__balance** to store the account balance.
 - Public Attribute: **owner**, which stores the name of the user.
 - Public Methods:
 - check_balance**: Displays the current balance.
 - deposit**: Adds a specific amount to the balance.
2. Allows the user to interact with the account (check balance, deposit money, exit) through a menu-driven loop.